

Techno-Economic Modeling of Hybrid PV/Wind/Biomass Microgrid Renewable Energy Systems Using Load Following Dispatch Method

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Abstract. This research presents a technical and economic assessment of a hybrid energy system for electricity generation, accessibility, sustainability and consumption in rural and semi urban locations in Nigeria. The aim is to determine the sizes, technical and economic considerations of the hybrid microgrid renewable energy infrastructure that could be suitable for 40 rural and semi-urban locations selected to cut across the Federal Capital Territory (FCT) and all the 36 states in the country. The cost of the components of the hybrid system and the energy generated per renewable energy (RE) source from the microgrid are determined for all the locations. The projected yearly electricity generated for each location for the hybrid system were determined using the hybrid optimization of multiple electric renewables (HOMER) energy modelling software for a period of 10 years from 2024 to 2033. The World Bank population growth rate of 2.5% was used to estimate the population each year and the associated load demand. Each location was assumed to have a minimum electric load of 0.76 kWh per day per person. To simulate long-term continuous implementation of the hybrid system, average solar irradiation, wind speed and available biomass resources for the selected locations were used. The mean annual wind speed ranged from 3.45 to 7.15 m/s. The solar radiation ranged from 4.43 to 6.24 kWh/m²/day. The per capita net present cost (NPC) ranged from 776.37 to 4130.21 USD per kWh, while the cost of energy (COE) ranged from 0.00196 to 0.0231 USD /kWh, respectively for the period. The results show that Nigeria as a country has ample renewable energy resource availability to meet minimum electric power demand for the locations considered. With a strong political determination, optimal utilization of these renewable resources (solar, wind and biomass) can be actualized. Researchers, Industrialists, Policy Makers and the Nigerian government should therefore take advantage of these abundant renewable energy resources in the country to develop a sustainable energy generation and consumption plan through its maximum utilization.

Introduction

Energy and power are required in nearly every sector of a country's economy. Adequate and sustainable energy generation promotes socio-economic development. Adequate power generation and supply will enhance many other benefits, including lower product costs, improved living standards, convenience at home and work, better perishables storage, a higher standard of education, etc. [1]. Also, there is a proven case of the possibility of eradicating poverty as a result of improved energy access, as shown by the China example (lifting a significant portion of its population out of poverty) [2].

With a population of over 206.14 million in 2020 and a yearly increase in population of 2.5%, Nigeria must effectively address its power issues if it is to reverse the epidemics of poverty, crime and low productivity [3]. Nigeria's national grid has a centralized conventional power generation and distribution that provides insufficient coverage throughout the country due to disparate distribution of crude oil and natural gas resources across geopolitical zones, among other challenges. Biofuels (firewood, charcoal etc.) and wastes (primarily agro-waste) account for approximately 80% of energy demand in rural areas, which are largely without any or sufficient electricity supply [4]. Consequently, reaching these un-served areas in Nigeria will take years since it is economically intensive and unattractive to expand the national grid to the rural vicinities with low energy demand. Investigating distributed energy supply is necessary to achieve broad access to energy by 2030 [5]. It therefore necessitated several studies on standalone hybrid renewable energy (RE) technologies, which combine RE sources to generate power in a continuous, affordable, effective, sustainable, and clean manner [6]. To build generation stations that depend on RE sources, government must make significant investments due to the high cost of electricity production [7]. Researchers use software to simulate potential projects. Combining solar and wind energy (which fluctuates all through the day depending on the condition of the weather) is possible. Biogas can be easily stored and used as a backup energy source [8]. According to Amam et al [9], sustainable energy systems have a combined use of conventional and RE sources in contrast to conventional energy systems.

Methodology

A microgrid is defined in [10] as a subsystem consisting of generation and associated loads that uses local control to facilitate its connection and disconnection to/from with the main grid in order to maintain a standard service during disturbances without harming the integrity of the transmission grid. A recent review of the microgrid renewable energy technology may be found in [11].

This study evaluates the performance of an off-grid PV-wind-biomass-battery hybrid energy system within 40 locations in Nigeria with a potential for grid connection based on the individual population of the locations. These locations are shown in table 1 for the selected locations.

It takes into account the sizing, system requirements and technical considerations, response of the power system, reliability analysis-based stability and feasibility study, and dispatch strategy-based control. Information on the amount of sunlight, wind speed, and biomass derived from [12, 13, 14] is summarized in table 2.

For each of the study areas, the Hybrid Optimization of Multiple Electric Renewables (HOMER) software is used to model and optimize the system's operations to ensure technical, economic and environmental feasibility while providing a minimum possible net present cost NPC, cost of energy (COE) and operating costs. A summary of the HOMER simulation procedure is presented in figure 1.

For size optimization, modelling of hybrid energy system components is an important process to providing its performance under various conditions. Mathematical modelling of the proposed hybrid renewable energy source (HRES) components is explained below:

Energy demand based on the population of the location. According to [15, 16], the average Nigerian consumes around 0.38 kWh of electricity per day based on a per capita annual consumption of roughly 139 kWh. This translates to a relatively low daily energy consumption compared to other countries. This is due to the low installed generation and distribution capacity in the country, as well as frequent power outages in areas where grid supply has been extended. This work used twice of this data which is 0.76 kWh per day per capita for the computation of energy demand E_d for the population of the location for the given year, P_i . Thus we have

$$E_d = 0.76P_i \quad (1)$$

Table 1: The study locations

SN	State	Town	Latitude	Longitude	SN	State	Town	Latitude	Longitude
1	Abia	Eluoma	5.74	7.46	21	Katsina	Malumfashi	11.79	7.61
2	Adamawa	Michika	10.62	13.39	22	Kebbi	Yauri	10.99	4.52
3	Akwa Ibom	Ekpene-Ukim	4.9	8.03	23	Kogi	Iyah-Gbede	8.01	5.99
4	Anambra	Abagana	6.18	6.98	24	Kwara	Offa	8.13	4.71
5	Bauchi	Azare	11.67	10.2	25	Lagos	Agege	6.61	3.32
6	Bayelsa	Oloibiri	4.67	6.31	26	Nasarawa	Auta-Balefi	8.94	7.67
7	Benue	Abinsi	7.75	8.75	27	Niger	Kuta	9.87	6.7
8	Borno	Biu	10.61	12.19	28	Ogun	Isonyin	6.83	3.98
9	Cross River	Odukpani	5.08	8.34	29	Ondo	Ipogun	7.31	5.07
10	Delta	Agbara-Otor	5.53	6.06	30	Osun	Ilesa	7.61	4.7
11	Ebonyi	Nkaliki	6.27	8.23	31	Oyo	Ibarapa	7.4	3.23
12	Edo	Auchi	7.06	6.27	32	Plateau	Gindiri	9.59	9.22
13	Ekiti	Ikoru	7.83	5.03	33	Rivers	Abara	5.05	7.01
14	Enugu	Adaba	6.56	7.13	34	Sokoto	Binji	13.22	4.91
15	FCT Abuja	Duboyi	8.89	7.18	35	Taraba	Takum	7.25	9.97
16	Gombe	Billiri	9.86	11.21	36	Yobe	Gashua	12.86	11.03
17	Imo	Oguta	5.72	6.82	37	Zamfara	Bukkuyum	12.13	5.46
18	Jigawa	Ringim	12.28	9.47	38	Enugu	Nsukka	6.85	7.39
19	Kaduna	Kagoro	9.6	8.38	39	Ogun	Ijebu-Ode	6.83	3.91
20	Kano	Kabuga	11.99	8.47	40	Borno	Maiduguri	11.83	13.15

The population of any location for year $n + 1$ is calculated based on its population for the current year n using the expression

$$P_{l(n+1)} = P_{l(n)}(1 + r) \quad (2)$$

where r is the population growth rate.

Number of batteries needed. The Trojan Battery TROJAN SIND 04 2145Ah is selected for this work. It belongs to the range of industrial solar batteries of Trojan and are designed to support renewable energy systems with large daily loads where the batteries are used regularly. Therefore, the total number of batteries required, N_b is calculated as follows

$$N_b = \frac{T_{ed}}{B_{ic}} \quad (3)$$

where T_{ed} = total daily energy demand in Watt hour (Wh), B_{ic} = individual battery capacity in Wh. It is noted that energy stored in the battery, E_b in Watt hour is the product of the output voltage and the rating of the battery in ampere hours (Ah).

The power produced by the battery bank in Ampere hours is given by the expression

$$P_{bk} = \frac{T_{ed}}{V_{rb}} \quad (4)$$

where V_{rb} is the voltage rating of the battery.

Solar panels required. The choice of solar panels used in this work is the Schneider 680W N-type 144 cells-16BB monocrystalline solar panel. It is designed with a large size ($2278 \times 1134 \times 30\text{mm}$), achieves high power output, utilizes high transmission low iron tempered glass, and includes bypass diodes for protection. Additionally, it exhibits excellent low irradiance performance and is resistant to salt, mist and ammonia. The number of required solar panels is given as:

$$N_{sp} = \frac{T_{ed}/(h_{ps}A)}{W_{sp}} \quad (5)$$

Table 2: Solar, wind and biomass resources and load demand for the study locations

SN	A	B	C	D	E	SN	A	B	C	D	E
1	5.19	3.45	135892	6914	5255	21	6.2	7.09	123834	11920	9060
2	6.68	4.4	117921	5000	3800	22	6.22	5.95	102551	10981	8346
3	6.29	4.45	243523	7177	5455	23	5.68	4.39	148160	9929	7547
4	5.15	3.59	205538	9189	6984	24	5.71	5.09	161752	8919	6779
5	6.08	4.9	119428	10545	8015	25	4.43	5.99	8462738	20600	15656
6	4.59	4.31	151389	2500	1900	26	5.79	5.1	202180	15284	11616
7	5.32	4.42	131514	3795	2884	27	5.91	5.08	262187	7528	5721
8	6.24	6	120830	9545	7255	28	5.48	4.4	397158	2051	1559
9	5.01	4.45	252789	10433	7929	29	5.5	4.35	295103	9984	7588
10	5.45	4.4	865563	9879	7508	30	5.3	4.28	126065	10020	7616
11	5.32	4.61	283871	12837	9757	31	5.18	4.37	398175	11070	8414
12	5.31	4.01	732899	10860	8254	32	5.79	6.65	182465	10400	7904
13	5.4	3.91	268169	8386	6374	33	4.59	4.4	1779173	12546	9535
14	5.35	5.42	394764	6000	4560	34	6.24	7.15	131462	10158	7721
15	5.72	4.4	755184	3000	2280	35	5.92	5.09	128171	9979	7585
16	6.08	6.01	154290	9902	7526	36	6.24	6	121482	10392	7898
17	5.3	3.36	105621	10795	8205	37	6.24	6.78	125791	11633	8842
18	6.24	7.01	104722	12024	9139	38	4.5	5.11	200808	444100	337516
19	6.01	5.91	202144	7008	5327	39	5.15	3.95	101924	381000	299627
20	6.15	7.05	2441554	12235	9299	40	6.01	5.85	222827	822000	964440

Key: A = Annual mean solar irradiation, (kWh/m^2), B = Annual mean wind speed, (m/s),

C = Biomass resources, (kg/day), D = Population, E = Annual load demand, kWh.

Data for solar radiation, wind speed and biomass resources adapted from [12, 13, 14].

where h_{ps} = number of peak sunlight hours, A = area of each panel ($5m^2$) and W_{sp} = wattage of one solar panel (680 W). Other details about the panel are shown in row A of table 3.

Solar panel configuration. Grid-tied solar panels are connected in series primarily to increase the total voltage of the system, which is essential for compatibility with grid inverters. This configuration allows for:

1. **Voltage Increase:** This is crucial because grid inverters typically require a certain voltage range to function properly.
2. **Reduced Power Loss:** Higher voltage reduces the current required to transmit power, minimizing resistance losses in the wiring. This is particularly important for long cable runs between the solar panels and the inverter.
3. **Efficiency in Low Light:** Series connections can be more efficient in low-light conditions, as the higher voltage helps the system to operate more effectively.
4. **Cost Savings:** Using higher voltage can sometimes allow for the use of smaller gauge wires, potentially reducing installation costs.

Therefore, the number of solar panels required for each location R_{sp} is equal to the number of solar panels connected in series, N_{sp} for the hybrid power system of the location. This can be mathematically expressed as

$$R_{sp} = N_{sp} \quad (6)$$

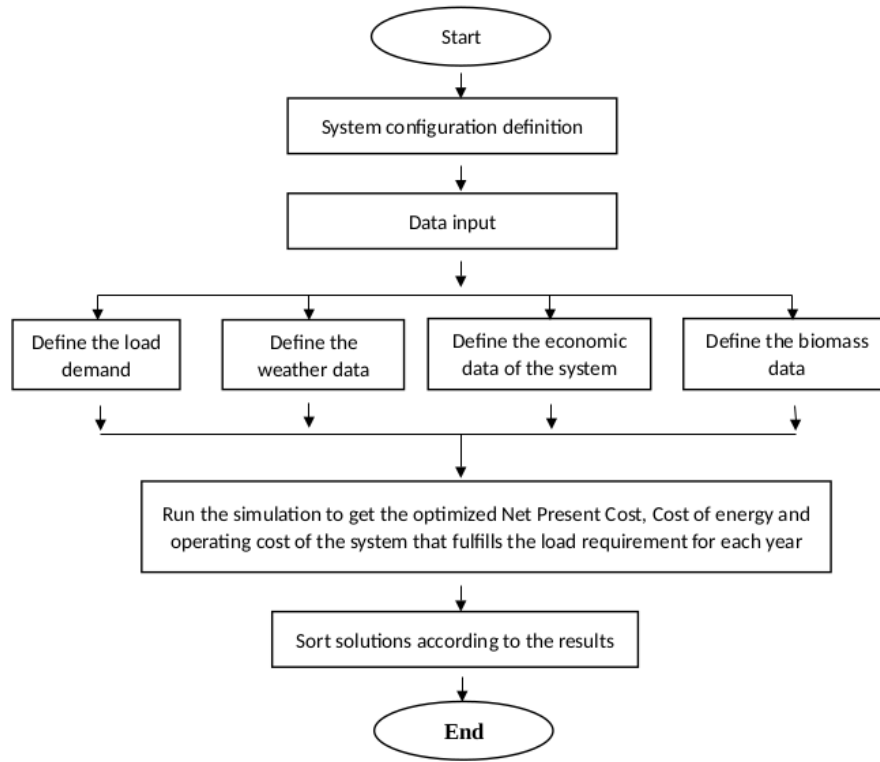


Fig. 1: HOMER simulation flowchart

The PV power output is then given by

$$P_{out} = \beta R_{sp} \delta \varepsilon_p \quad (7)$$

where, β = wattage of a panel, δ = number of sunlight hours, and ε_p = efficiency of the panel.

Capacity of the inverters. The choice of inverter used in this work is Eaton's Power Xpert-Storage grid-tie inverter that offers reliable high-performance power conversion for utility-scale grid-connected battery applications (see row D of table 3). High performance is realized with a wide end-of-charge to full-charge voltage window, which allows full battery storage potential to be achieved. Its conservatively rated magnetics, liquid-cooled power modules and a rugged enclosure suitable for the toughest environments make it one of the most dependable energy storage grid-tie inverter in the market. Therefore, the capacity of the inverters in kVA is expressed as:

$$C_{inv} = \frac{P_{tot}}{p_f} \quad (8)$$

where P_{tot} is the total wattage of the system in kW, and p_f is the power factor (0.8 in the present case).

Number of inverters. The number of inverters, N_{inv} , that can cater for the load demand of each location is expressed as the total capacity of required inverters divided by the capacity of one inverter. Thus,

$$N_{inv} = \frac{C_{inv}}{C_{inv1}} \quad (9)$$

where C_{inv1} is the capacity of one inverter (2.25 kW for the Eaton Xpert inverter employed).

Charge controller The capacity of a charge controller required for a solar system is calculated by dividing the total wattage of the solar panels by the voltage of the battery. A safety factor of 1.25 is

Table 3: System architecture of the hybrid PV, wind and biomass system

Component	Capacity (kW)	Capital (\$)	Replace-ment (\$)	Ground reflectance (%)	Hub height (m)
A	680	4,262/unit	4,262/unit	20	-
B	100	150,000/unit	150,000/unit	-	24
C	Varies per location	Varies per location	Varies per location	-	-
D	2250	61,145/unit	61,145/unit	-	-
E	8580	3,957/unit	3,957/unit	-	-
F	80-amp 24V	195/unit	195/unit	-	-

Component	Derating factor (%)	Inverter input (Lifeline (years))	Inverter input (Efficiency (%))	Rectifier input (Relative capacity (%))	Rectifier input (Efficiency (%))
A	80	-	-	-	-
B	-	20	-	-	-
C	-	-	-	-	-
D	-	10	97	100	97
E	-	15	-	-	-
F	-	-	-	-	-

Key: A = Solar paanel (Schneider 680W N-type 16BB), B = Wind turbine (Norvento nED 100-22), C = Biogas generator, D = Eaton's Power Xpert inverter, E = Battery (TROJAN SIND 04 2145Ah), F = Charge controller

applied to account for variable power outputs and ensure sufficient capacity for the controller. This is expressed mathematically as

$$C_c = 1.25 \frac{T_{wat}}{B_{volt}} \quad (10)$$

where T_{wat} is the total wattage of the solar panels, and B_{volt} is the battery voltage.

Number of charge controllers The capacity of the charge controller chosen for this work is 80 amp 24 volts (see row F of table 3). Therefore, the number of charge controllers can be expressed as follows:

$$N_{inv} = \frac{C_c}{80 \times 24} \quad (11)$$

Wind electric power generation. Wind electric power generation systems are widely reported in the litterature. A good discussion may be found in [17, 18, 19, 20, 21, 22, 23]. The wind turbine used for this study is a Norvento nED 100-22 designed for industrial and commercial self-use energy generation. Further details are shown in row B of table 3. It is a 100kW wind turbine with 22m diameter, $380m^2$ swept area, 400 Volts (V) rating, 24.5m hub height, cut-in speed of 3 m/s, cut-out wind speed of 20 m/s, survival wind speed of 59.5m/s, and wind Power Density of $263.2W/m^2$. It is suitable for grid connections using its electronic power converter. It has "low mode" versions for reduction of noise pollution in sensitive areas and only requires once a year check for maintenance. Wind potential becomes more apparent when wind power is calculated using the potential wind speed (preferably

mean speed) at the height of interest to determine whether it meets the demand to rotate a turbine to an acceptable rotational speed and torque.

Mean wind power density. The mean density of wind power (P_d), refers to the quantity of energy available at a given location for conversion by a wind turbine. It is usually measured in watts per square meter. As a result, it is an extremely useful method for determining the wind resource available at a prospective site. It can be calculated with the following equation:

$$P_d = \frac{1}{2} \rho A_w V_w^3 \quad (12)$$

where, ρ is the density of air (1.225 kg/m^3), V_w is the mean wind velocity (m/s), and A_w is the cross-sectional area of the wind turbine (m^2).

Wind energy potential. This is the power derived from a wind turbine (in Watts) and it is expressed as

$$P_w = 0.5 C_p \rho \pi R_{bld}^2 V_w^3 \quad (13)$$

where, C_p is the coefficient of performance (efficiency factor), ρ is the air density (kg/m^3), R_{bld} is the blade length (metres) and V_w is the wind speed (m/s).

Number of wind turbines required. This is the needed number of wind turbines for the hybrid system in each location and it is expressed as follows:

$$N_{wt} = \frac{P_w}{P_{wt1}} \quad (14)$$

where P_{wt1} is the capacity of one wind turbine in Watts.

Biomass system model. Many biomass hybrid or non-hybrid systems for electricity generation have been reported in the literature. See for example [24, 25, 26, 27, 28]. The choice of biogas generating plant for this work is based on the biomass resources conversion efficiency of 40%, 10 hours of daily use at the location and the energy demand of the respective locations. The capacity of the biomass generating set is given mathematically as:

$$B_{gcap} = B_{res} C_{eff} G_{hr} E_{btu} \quad (15)$$

where B_{res} is the available biomass resources in tons/day, C_{eff} is the conversion efficiency of the biomass generating set, G_{hr} is the number of operating hours per day, and E_{btu} is the energy demand in kWh

Economic viability of the hybrid system. For each proposed location, the economic viability of the hybrid solar-wind-biomass power system is evaluated using net present cost (NPC), annual life-cycle savings (ALCS), cumulative cash flows, simple payback and equity payback periods, and simple cash flows.

The net present cost, (NPC), is the total value of all future cash flows, discounted in today's currency at the current discount rate. It is determined using the expression

$$NPC = \sum_{n=0}^N \frac{\bar{C}_n}{(1+r)^n} \quad (16)$$

where N = the project life in years, r = the discount rate, $\bar{C}_n$ = the after-tax cash flow for the year n , given by:

$$\bar{C}_n = C_n - T_n \quad (17)$$

where C_n = annual net cash flow for the year n ; and T_n = the tax value for year n .

Annual life-cycle savings, ALCS. This is calculated by the HOMER modelling software. It is the levelized nominal yearly savings with the same life and net present cost as the project. It is provided by the equation

$$ALCS = \frac{NPC}{(r^{-1})} \quad (18)$$

where all variables have already been defined.

Simple payback, SP (Years). This is the period that it takes for the projects' earnings to cover the initial investment. The earnings here refer to the net benefit the project generates each year either through the amount of money saved from not paying electricity bills or revenue made from selling electricity. This can be expressed as:

$$SP = \frac{C_{tii}}{C_{asr}} \quad (19)$$

where, C_{tii} is the total initial investment which is the cost of all the system components and installation, C_{asr} is the annual savings, which is the excess electricity generated in the given location multiplied by the cost of electricity for the given location. Now, in areas covered by the Enugu Electricity Distribution Company (EEDC), the current cost of electricity is NGN 53.38 per kWh. Using an exchange rate of 1 USD = NGN 1500, it follows that the cost of electricity is $53.38/1500 = 0.036$ USD. This rate is applied throughout the country for convenience.

Levelized cost of energy (LCOE). This means dividing the total cost of building and operating the power plant over its lifetime by the total amount of electricity it generates during that time. This is given by the expression

$$LCOE = \frac{T_{cl}}{T_{ep}} \quad (20)$$

where T_{cl} is the total cost of building and operating the power plant divided by the lifetime, N_{lft} ; and T_{ep} is the total electrical energy produced divided by the lifetime, N_{lft} .

Operating cost of the hybrid system. The operating cost of the hybrid system is the annualized value of all costs and revenues other than initial capital costs. This is calculated from the expression

$$C_{oper} = \frac{C_{fuel} + C_{maint} + C_{other}}{E_{tot}} \quad (21)$$

where: C_{fuel} is the total cost of fuel, C_{maint} is the total cost of maintenance, and C_{other} is other expenses such as insurance, labour, administration, etc. All the costs are in (US \$/year)

Results and Discussion

The study is applied to 40 different rural or semi-urban locations in Nigeria as shown in table 1. They are selected to cover the Federal Capital Territory (FCT), Abuja, one location in each of the 36 states and three other towns.

The mean annual solar radiation data, wind speed and biomass availability for each location were obtained from the literature as shown in table 2. The projected load demand is also shown.

For each location, a suitable mix of PV, wind and biomass power plants are chosen from the generic values presented in table 3. The minimum, mean and maximum values of the resulting PV systems is shown in table 4.

Similarly, the minimum, mean and maximum values of the resulting wind electric systems is shown in table 5.

From this, the capacity of the PV, wind and biomass systems were determined for all the locations as shown in table 6. The number of each component and other details for each location are presented

Table 4: Range of values of the PV systems employed

Item	Num- ber of bat- teries (trojan batter- ies)	Num- ber of Series con- nection	Num- ber of solar panels re- quired (schnei- der electric 680W each)	Series con- nection solar panels	Capac- ity of the in- verters (kVA)	Num- ber of invert- ers	Ca- pacity of the charge con- trollers (Amp)	Num- ber of charge con- trollers
Min	182	182	1	1	4207	3	213	3
Mean	4427	4427	12	12	49123	25	2482	32
Max	72812	72812	183	183	769973	343	38888	487

Table 5: Range of values of the wind-electric systems employed

Item	Wind speed, (m/s)	Wind Energy Potential/ Wind power Output (Watts)	Number of Turbines required
Min	1.225	353422	4
Mean	2	1387234	15
Max	1.225	3405604	35

in Ezeanyanwu [29]. The capacities indicated are sufficient to take care of the electric power needs of the locations indicated.

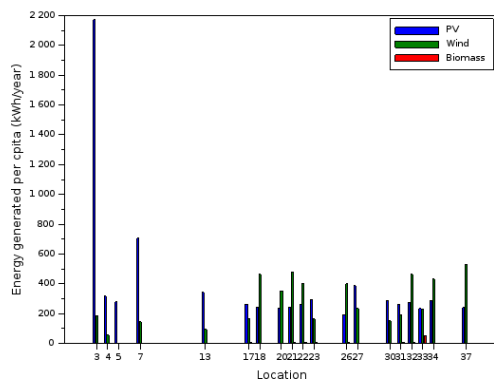
In order to make the presentation of other details more legible, an algorithm was developed based on the *perctl* function in the SCILAB computing software to select data points for the percentile values of 20, 30, 40, 50, 60, 70, and 80.

The projected energy outcome were determined for each year for a period of ten years for the three energy sources considered. For convenience, only the per capita energy generated from each of the sources for the first year is shown in figure 2. Part (a) of the figure shows the per capita energy generated from PV, wind and biomass, while part (b) shows the per capita energy generated, consumed and the excess available which could be fed to the grid. The minimum, mean and maximum values of each energy source is shown in columns 2 to 4 of table 7. It is seen that the mean value for the photovoltaic system is highest compared with the wind and biomass systems. For most of the sites, the PV system also provides the largest amount of the power per capita. Further, it is seen that the energy generated varies from 375.05 to 2358.21 kWh per capita per annum, the mean value being 764.31 kWh. On the other hand, the energy consumed is constant at about 277 kWh, which, as expected, is the product of the average per capita consumption of 0.76 kWh multiplied by 365 days in the year. The per capita excess power produced varied from 70.61 to 2054.59 kWh per annum, the average value being 470.57 kWh. Thus, every location has the capacity to produce excess power for grid supply.

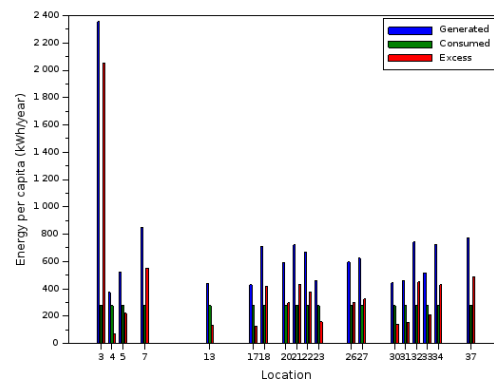
The payback period for the locations considered varied based on the resources available. The minimum payback of 2 years was obtained for Ijebu Ode in Ogun State (location number 39), while

Table 6: Total capacity of the hybrid system for all the locations

SN	PV	Wind	Biomass	Total	SN	PV	Wind	Biomass	Total
	kVA	kVA	kVA	kVA		kVA	kVA	kVA	kVA
1	8415	307	671	9393	21	12622	2657	1053	16332
2	8415	635	421	9471	22	12623	1571	804	14998
3	8415	657	1247	10319	23	12622	631	1049	14302
4	12622	345	1347	14314	24	8415	983	1029	10427
5	12623	877	899	14399	25	21038	1602	124260	146900
6	4208	597	271	5076	26	16830	989	2203	20022
7	4207	644	356	5207	27	8415	978	1407	10800
8	12623	1610	823	15056	28	4208	653	581	5442
9	12622	657	1880	15159	29	12623	614	2101	15338
10	12623	640	6096	19359	30	12623	585	901	14109
11	12623	731	2598	15952	31	12623	623	3143	16389
12	12622	481	5674	18777	32	12623	2192	1353	16168
13	8415	446	1604	10465	33	12623	657	15910	29190
14	8415	1187	1688	11290	34	12623	2725	952	16300
15	4208	649	1616	6473	35	12623	983	912	14518
16	12623	1619	6021	20263	36	12623	1610	900	15133
17	12622	283	813	13718	37	12623	2324	1044	15991
18	12623	2568	898	16089	38	420750	995	79456	485310
19	8414	1539	1011	10964	39	374468	460	34600	409528
20	12623	2612	21293	36528	40	1194930	1493	163195	1359618



(a) Generated from PV, wind, biomass



(b) Energy generated, consumed and excess

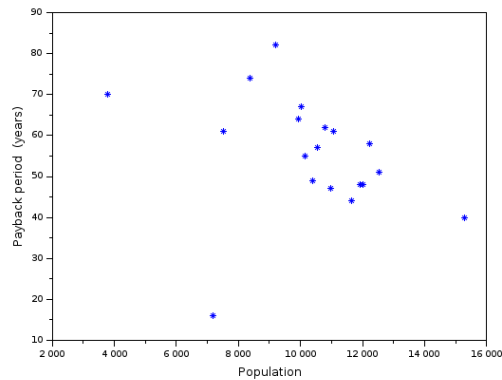
Fig. 2: Projected energy per capita

the maximum payback period was 82 years for Abagana in Anambra State (location 4). The payback period generally decreased with increasing population as shown in figure 3(a).

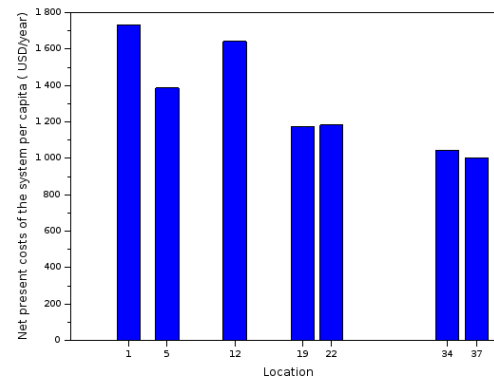
The net present costs of the system is shown in figure 3(b). These costs varied from a minimum of USD 776.37 to a maximum of 4130.21. The costs of energy of the system is shown in figure 4(a). This varies from 0.00196 to 0.0231 US cents per capita per annum, the average value being 0.00547. These costs are well within the income levels of even families considered to be poor.

Table 7: Minimum, mean and maximum of the output variables (kWh)

Description	PV	Wind	Biomass	Generated	Consumed	Excess
Minimum	139.99	0.2438	0	375.05	277.38	70.61
Mean	416.39	335.39	5.946	764.31	277.41	470.57
Maximum	2173.34	2027.18	65.51	2358.21	277.44	2054.49

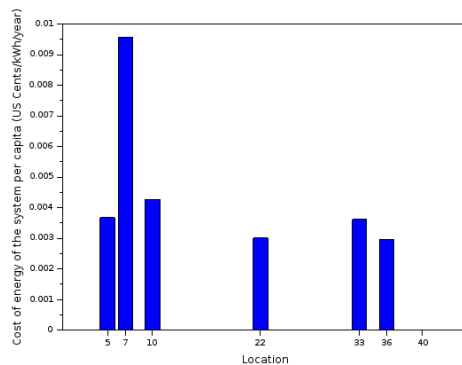


(a) Payback period

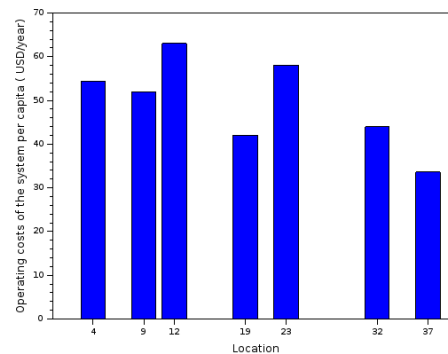


(b) Net present costs

Fig. 3: Projected payback period and net present costs of the system



(a) Energy costs



(b) Operating costs

Fig. 4: Cost of energy and operating costs of energy of the system

The operating costs of the system is shown in figure 4(b). This varies from 11.28 to 257.94 US cents per kWh per capita. This is certainly affordable for low income earners.

Conclusion

The renewable energy resource availability in forty (40) semi-urban and rural locations in Nigeria were considered. Using available technologies for photovoltaic conversion, wind electric power conversion and biomass electric power generation, the projected energy outcome for each location was determined using the HOMER software. The minimum, mean and maximum values of the per capita electricity produced from each renewable energy type was determined. The per capita electricity generated, consumed and the excess available for grid supply were obtained. The payback period was found

to vary between 2 and 82 years. The net present costs of the system varied between USD 776.37 to 4130.21 per capita. On the other hand, the operating costs varied between 11.28 to 257.94 US cents per kWh per capita. It is seen that each of the locations considered has the potential to produce excess power for possible grid supply. Further, the cost of the electricity produced is affordable even for low income earners. The findings in this study agree with some of the findings reported in [11].

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